CS4100+CS4110 — Computer System Design (+ lab)

Tier	B
Phase (order)	5
Do this first	CS2600, CS3500
Estimated effort	10-12 weeks
Unlocks	CS6886, CS6847

Roadmap

1. Building blocks: CPU, storage, I/O.
2. Performance specification and measurement.
3. Integration and distributed resources.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Livestream - Computer Architecture - ETH Zürich (Fall 2023) (32 videos). Onur Mutlu *Computer Architecture* (ETH Zurich, Fall 2023): memory systems, caches, prefetching, multicore.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Advanced Computer Architecture (34 videos). NPTEL *Advanced Computer Architecture* (IIT Guwahati).
- Advanced Computer Architecture (58 videos). Smruti Sarangi (IIT Delhi) *Advanced Computer Architecture*: he also wrote a free textbook.
- Spring 2015 -- Computer Architecture Lectures -- Carnegie Mellon (39 videos). CMU 18-447 Spring 2015 (Onur Mutlu): official CMU recording.

⚡ Short videos for intuition and revision

- Multi-Core Computer Architecture (49 videos). NPTEL *Multi-Core Computer Architecture* (IIT Guwahati).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–2	Intro + memory systems challenges.
2-3	9, 12–14, 31–32	Memory latency, memory controllers, prefetching, virtual memory, cache design. Write a cache simulator (or use gem5).

Week	Lectures	What to do
4	18–21	Parallelism, multiprocessors, cache coherence, interconnects.
5	27–30	VLIW, systolic arrays (TPU-style), SIMD and GPU architecture, GPU programming.
6	3, 5, 15–17	Processing-in-memory, emerging memories, flash/SSDs.
7	22, 26	Simulation and on-chip networks. Project: extend your RISC-V CPU (CS2600 lab) with a cache and branch predictor.

▼ **All lectures in *Livestream - Computer Architecture - ETH Zürich (Fall 2023)* (32 videos)**

1. Computer Architecture - Lecture 1: Introduction and Basics (Fall 2023)
2. Computer Architecture - Lecture 2: Memory Systems: Challenges and Opportunities (Fall 2023)
3. Computer Architecture - Lecture 3: Processing near Memory (Fall 2023)
4. Computer Architecture - Lecture 4: Programming a Real-world PIM Arch. and Enabling PIM (Fall 2023)
5. Computer Architecture - Lecture 5: Processing using Memory (Fall 2023)
6. Computer Architecture - Lecture 6: Memory Security, Reliability Problems and Solutions (Fall 2023)
7. Computer Architecture - Lecture 7: Cutting Edge Research on DRAM Read Disturbance (Fall 2023)
8. Computer Architecture - Lecture 8: Data Retention and Memory Refresh (Fall 2023)
9. Computer Architecture - Lecture 9: Memory Latency (Fall 2023)
10. Computer Architecture - Lecture 10: Genome Analysis (Fall 2023)
11. Computer Architecture - Lecture 11: Cutting-Edge Research in Computer Architecture (Fall 2023)
12. Computer Architecture - Lecture 12: Memory Controllers (Fall 2023)
13. Computer Architecture - Lecture 13: Memory Controllers II & Prefetching (Fall 2023)
14. Computer Architecture - Lecture 14: Prefetching II (Fall 2023)
15. Computer Architecture - Lecture 15: Emerging Memory Technologies (Fall 2023)
16. Computer Architecture - Lecture 16: Flash Memory and Solid-State Drives (Fall 2023)
17. Computer Architecture - Lecture 17: Flash Memory and Solid-State Drives II (Fall 2023)
18. Computer Architecture - Lecture 18: Parallelism & Heterogeneity (Fall 2023)
19. Computer Architecture - Lecture 19: Multiprocessors and Memory Ordering (Fall 2023)
20. Computer Architecture - Lecture 20: Cache Coherence (Fall 2023)
21. Computer Architecture - Lecture 21: Interconnects (Fall 2023)
22. Computer Architecture - Lecture 22: Simulation (Fall 2023)
23. Computer Architecture - Lecture 23: Cutting-Edge Research in Computer Architecture (Fall 2023)
24. Computer Architecture - Lecture 24: Discussion Session (Fall 2023)
25. Computer Architecture - Lecture 25: Cutting-Edge Research in Computer Architecture (Fall 2023)
26. Computer Architecture - Lecture 26: On-Chip Networks (Fall 2023)
27. Computer Architecture - Lecture 27: VLIW Architectures (Fall 2023)
28. Computer Architecture - Lecture 28: Systolic Array Architectures (Fall 2023)
29. Computer Architecture - Lecture 29: SIMD & GPU Architectures (Fall 2023)
30. Computer Architecture - Lecture 30: GPU Programming (Fall 2023)

31. Computer Architecture - Lecture 31: Virtual Memory (Fall 2023)
32. Computer Architecture - Lecture 32: Cache Design and Management (Fall 2023)

Books, notes and websites

- Book: Wolf, *Computers as Components*
- Book: Heuring & Jordan, *Computer Systems Design and Architecture*
- Book: Bryant & O'Hallaron, *CS:APP*; CMU 15-213 lectures

Practice / proof of understanding

Profile a program with `perf`, find the bottleneck, and improve it by 2x.

What your IITM course sheet says

From the PDF: CS4100

Official description: Basic principles of System Design: system components in simple system; Workstations, servers. Building blocks for Computer Systems; CPU, Storage, I/O, Multimedia devices The hardware-software interface, Hardware features influenced by software requirements. Specifications of the performance of a system. How to effectively utilize the strength of a component to get improvement in performance? Testing systems for performance. Computing with distributed resources. Basic principles of system integration, when building larger systems.

Official textbooks: 1. Wayne Hendrix Wolf, "Computers as Components: Principles of Embedded Computer Systems Design", Academic Press 2001. 2. Vincent P. Heuring and Harry F. Jordan, "Computer Systems Design and Architecture", Pearson Education Asia, 2001.

From the PDF: CS4110

Official description: Students are expected to design modules and sub-systems which may be put together to form computers. Several experiments are included for software design for computer systems. These will be based on workstations and development systems working in a networked environment. The laboratory experiments will supplement the lectures given in the course CS 410.

Official textbooks: NIL